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WarpForm: An Accessible Loom Add-On for Weaving 3D and Non-Rectangular Textile-forms

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Textile-form Weaving, in which the fabric and form are created simultaneously on the loom, can support zero-waste, on-demand production of textile-based products. However, textile researchers and designers are limited in their ability to explore the full potential of Textile-form Weaving because most looms do not allow for all variables to be manipulated. In this paper, we present WarpForm, an accessible add-on for a standard hand-operated loom that opens up new possibilities for weavers. Specifically, through enabling the variable consumption of individual warp threads, WarpForm allows weavers to create 3D and non-rectangular forms directly on the loom. Through a description of the add-on, the weaving process, woven samples, and a demonstrator, this paper aims to provide textile researchers and designers with accessible knowledge and tools to uncover further potential in Textile-form Weaving, moving the textile industry towards more sustainable approaches.

Keywords: Textile-form Weaving; Textile Design, Fabrication tools, Loom add-on; Zero-waste; Variable Warp Consumption

1. Introduction

Textiles provide us with warmth, comfort, and protection, but also significantly contribute to global emissions, pollution, resource depletion, and social issues (Niinimäki et al., 2020a). The textile industry's negative impacts are in part due to its wasteful production methods. Most textile products are made in a subtractive manufacturing approach; large prefabricated sheets of fabric are cut and sewn into forms, a process whereby an average 25%-35% of the raw materials is lost (Kerr & Landry, 2017; Reverse Resources, 2017).

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Textile-forms, on the other hand, allow the additive manufacture of textile objects. Understood as the simultaneous creation of textile and form, Textile-forms can contribute to a more sustainable textile industry through zero-waste and on-demand production (McQuillan & Karana, 2023). Knitting lends itself well to the production of textile-forms, because knitted fabric can be built up non-uniformly (Rob Thompson, 2014). Integral knitting (commonly known as 3D knitting) is therefore already used on an industrial scale for the low-waste production of garments and shoes (e.g., Uniqlo 3D knit, Nike Flyknit, Adidas Primeknit). Textile-form Weaving (TfW); which could be seen as the weaving counterpart of 3D knitting, however, is rare in industry, as the majority of both hand-operated looms and industrial looms are designed to weave flat, rectangular fabrics. Most looms have pre-installed warp threads with uniform yarn consumption across the width and length of the fabric, leaving little flexibility for shaping during weaving. Innovation and efforts towards sustainable fabrication, therefore, primarily focus on efficiency, speed, and the development of materials (fibre and yarn) (Seyam, 2023). Researchers and designers have explored how looms designed for flat weaving can be utilized to create Textile-forms with the help of multi-layer structures (McQuillan, 2020; Piper & Townsend, 2015; Shi et al., 2024; Voorwinden & McQuillan, 2025). These examples show that it is possible to create entire garments directly on the loom; however, these researchers have also noted significant limitations they encountered when working on looms designed for flat fabric.

In textile engineering, 3D woven textiles are typically classified as solid, hollow, shell, and nodal, and can be produced on a conventional 2D loom or specialised 3D looms (Chen et al., 2011; Unal, 2012). In this section, we focus on the creation of shell-woven fabrics. These are single-layered fabrics that extend out of the X-Y fabric plane, directly on the loom (Fig. 1). Shell woven fabrics can be created through manipulating three primary spatial variables:

- The spacing between adjacent warp or weft threads.
- The placement of interlacement points along the Z-axis.
- The incorporation of additional warp or weft threads in specific zones.

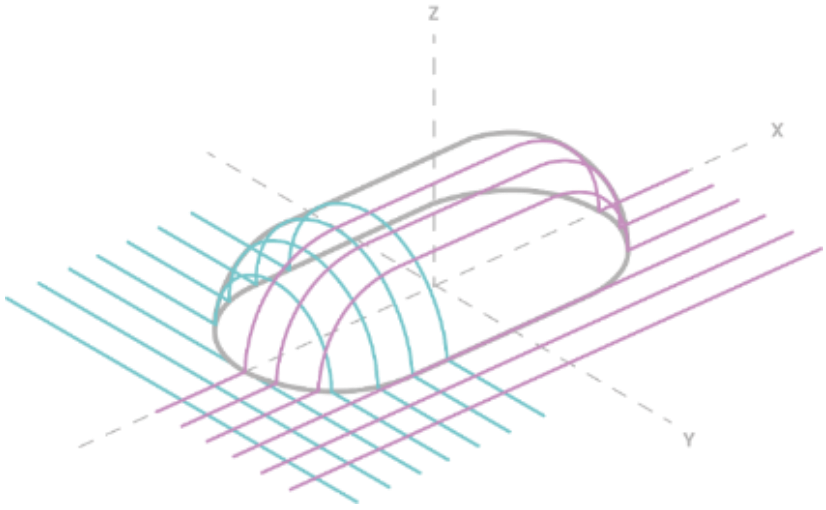


Figure 1: Schematic showing the additional warp (pink) and weft (blue) required to travel along the curvature of a 3D shape.

To create shell-woven textiles on conventional 2D looms, researchers have explored various methods for creating shape; for example, adjustable reeds to change the curvature of the weft during beat up (Cnaani & Sterman, 2026), the crimp of different weave structures to compress and extend the spacing of the structure (Busgen, 1999; Chen & Tayyar, 2003), partial weaving with shuttles to add supplementary material in particular weaving zones (short-row weaving) (Chen et al., 2024), and multilayer weaving, in which the weft only travels through part of the warp (Busgen, 1999). To curve the weft out of the fabric plane during beat-up, 3D moulds made from slotted fins and adjustable lamellae have been developed (Koranne, 2014; Kühl, 2013).

One challenge in creating shell-woven fabric with conventional 2D looms is the overall fabric take-up. 3D shell geometry requires threads to traverse different arc lengths (Fig. 1), so individual yarns must be consumed at different rates. Traditional looms are limited by uniform take-up and let-off, hence enforcing planarity. Multiple warp beams or creel systems enable more control over the let-off of warp threads, but varied take-up is a greater challenge. Researchers have developed specialised hardware, such as shaped fabric beams (Koppelman, 1964). In these methods, the fabric is kept flat on the loom, relying on the tension in or the curvature of the threads to pull it into a 3D form once taken off the loom. Therefore, the available forms are limited by the properties of the warp and weft materials, such as elasticity. Craft practitioners present methods for manually pulling forth and fixating areas of the fabric during the weaving process on a hand-operated loom (Eyring, 2020). Such craft approaches require extensive manual manipulation or hand-implemented post-processing, which is not compatible with today's industrial textile industry.

Later advancements resolve these limitations with shape-independent take up systems at or close to the fell, such as individually driven take-up rollers (Busgen, 1999), weft retention hooks (Dominik et al., 2024), or a fabric clamp (Chen et al., 2024). These systems maintain tension on the warp in the fabric-forming zone (between the warp let-off and the fell) while allowing the woven fabric beyond this area to deform freely according to its local crimp requirements and properties.

While these researchers have made interesting strides in developing shell-woven textile-forms, their approaches lack accessibility and versatility. Industrial-oriented approaches have created 3D and non-rectangular forms through dedicated machinery that are often too specialised, complex, and expensive, making them difficult to access or replicate for textile designers and researchers. Craft practitioners have explored shell-woven textile-forms through manual approaches that require a high level of skill, manual post-processing steps, and often rely on an individual’s own methods and lo-fi tools. Overall, there is a lack of accessible, versatile loom setups that allow weavers to explore weaving beyond conventional 2D planes, and as a consequence, few examples of TfW that explore shell weaving, and a lack of knowledge for how these could be designed. This lack of tools, examples, and knowledge creates a barrier for textile researchers and designers to engage directly with material processes, limiting their distinctive potential to uncover new insights into Textile-form Weaving through hands-on experimentation and exploration (Barati & Karana, 2019).

In this paper, we present WarpForm (Fig. 2), an add-on for a standard dobby loom that allows weavers to create 3D and non-rectangular Woven Textile-forms directly on the loom. To demonstrate

WarpForm's form-giving capabilities, a collection of samples is presented, alongside a step-by-step guide on how to make form by means of short-row weaving. WarpForm's potential for research in textile fields is further illustrated through a prototype of a near-net-shape 3D shoe upper.

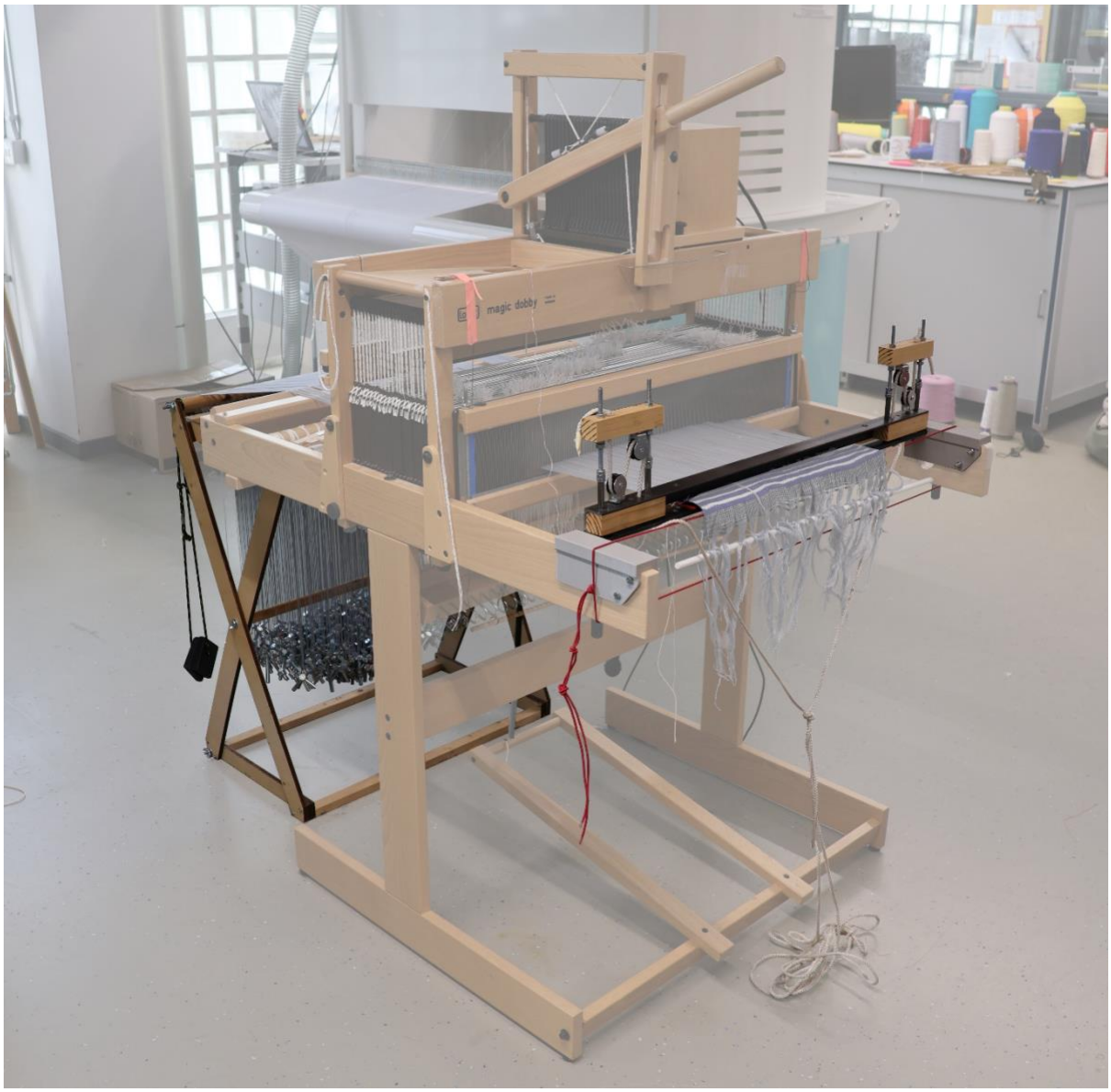


Figure 2: Picture of the final WarpForm prototype installed on a Magic Dobby Loom with the add-on parts highlighted.

2. Our Approach

This research was carried out by a multidisciplinary team of product designers and textile experts. Although new to weaving, product designers Savanne Klop and Manuel Borrego Carazo applied industrial design engineering expertise to the design and development of functional prototypes, while textile experts Milou Voorwinden and Holly McQuillan contributed their deep understanding of the textile industry, weaving processes and textile-form design.

The team built on insights gained from our earlier research (Voorwinden et al., 2026) that developed a toolkit to introduce weavers to unconventional form-making on a modified frame loom with similar variables available - primarily relating to warp left off and fabric take up. The samples produced on this simple frame loom were promising, encouraging us to explore more complex shapes. However, the low density and the time required to work on the frame loom led us to seek tools better suited to advanced material research and prototyping.

WarpForm's physical add-on and knowledge resource were developed through an iterative Research through Design (Stappers & Giaccardi, 2017) approach, guided by functionality and accessibility requirements as outlined below.

From a **functionality** perspective, WarpForm must enable additional material to be added to specific zones of the fabric by inserting partial weft rows (Fig. 3).

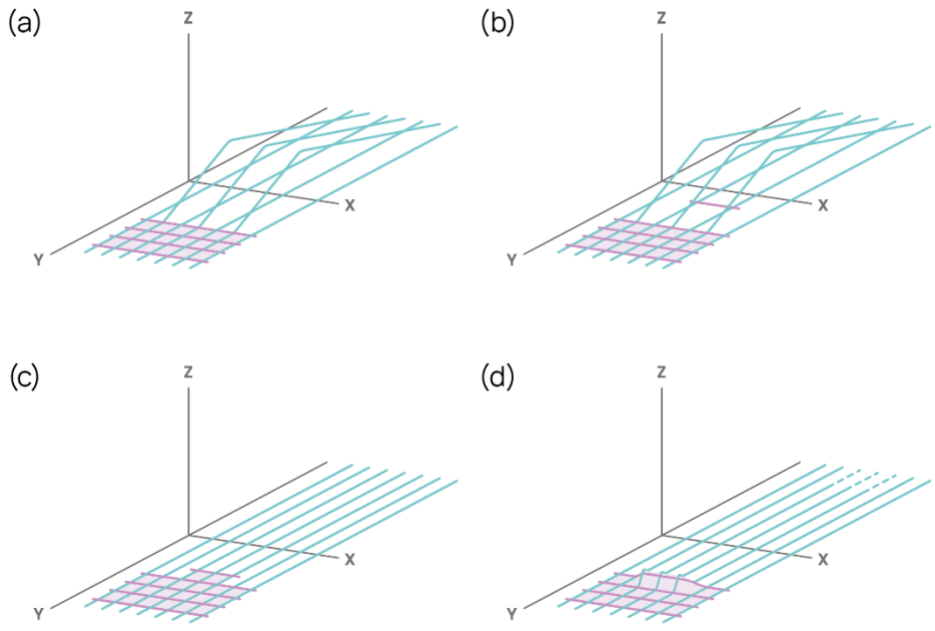


Figure 3: Schematic showing the weaving process that WarpForm must facilitate. (a) The warp is shed. (b) A partial weft row is inserted. (c) The warp threads are lowered, and the weft is beaten up against the fell. (d) The partial weft row and the interlaced warp threads are moved forward until the fell is aligned, causing the additional material to be pushed up, out of the fabric plane.

To facilitate this, WarpForm must enable three core functionalities when used in combination with a creel system (Fig. 4):

1. WarpForm must allow individual warp threads to be pulled forward at different rates.
2. WarpForm must enable the tension in each warp thread to be maintained in the fabric-forming zone.

3. WarpForm must fixate the warp at the fell, leaving the woven fabric untensioned.

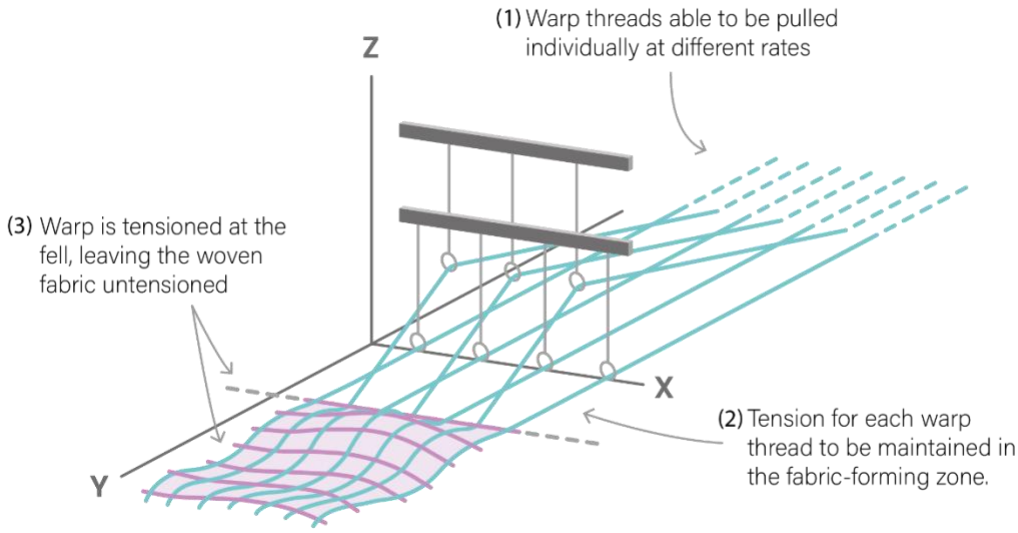


Figure 4: Schematic of the loom showing the three core functionalities WarpForm must enable.

To ensure **accessibility**, we designed WarpForm and the supporting information with the following requirements.

- WarpForm must be a non-permanent attachment for a standard dobby loom and require no permanent alterations to the loom.
- WarpForm must be constructable from parts and materials available at a standard hardware store, and custom-made parts must be producible with machinery and tools available in a prototyping makerspace (e.g., 3D printers and laser cutters).
- WarpForm must be a simple, inexpensive, and mechanical tool.
- The weaving process must be understandable and replicable for both beginner and experienced weavers.

The requirements were integrated into WarpForm's design over three phases of development in which methods for keeping the fell in place, creel systems, and different looms were explored. The final prototype was built for a 24-shaft Louët Magic Dobby loom.

During WarpForm's development, to understand the potential of this tool to create varying rounded shapes, sharp corners, and curved fabrics, a selection of samples were woven. Although form can also be created by varying the weave structure within and between weft rows, we focused this research on short-row weaving in a plain weave 1x1 structure. This was done because building up a form by

adding extra material row by row is simpler and perhaps more intuitive - and therefore a better entry point for first-time users. Furthermore, short-row weaving is comparable to short-row knitting, so existing knitting resources and patterns can serve as inspiration or guides for experimentation in weaving.

Observations made during weaving influenced subsequent samples. Photos, videos, sketches, and written observations were produced during the weaving process. The placement of material in each sample was described in a weaving plan, and the resulting samples were photographed after being taken off the loom. The samples were clustered to provide an overview of WarpForm's capabilities.

A detailed walkthrough of the weaving process was developed based on our exploration and sampling, to serve as a guide for other weavers. Finally, we applied the insights gained from this exploration to a near-net-shape 3D shoe upper, a domain chosen for its form-sensitivity and requirements.

3. Development and Design of WarpForm

WarpForm was developed through an iterative design process (Fig. 5). In the first iteration, small-scale prototypes tested both a hook system and a clamp system to determine the optimal method for fixing the warp at the fell. Although the hooks were more reliable, we opted for the clamp for its ease of use. In the second iteration, a clamp for a table loom was created and tested in combination with a simple creel system using weights. Lastly, the clamp and creel system was scaled up to fit a dobby loom and equipped with a foot-activated pulley system to open and close the clamp.

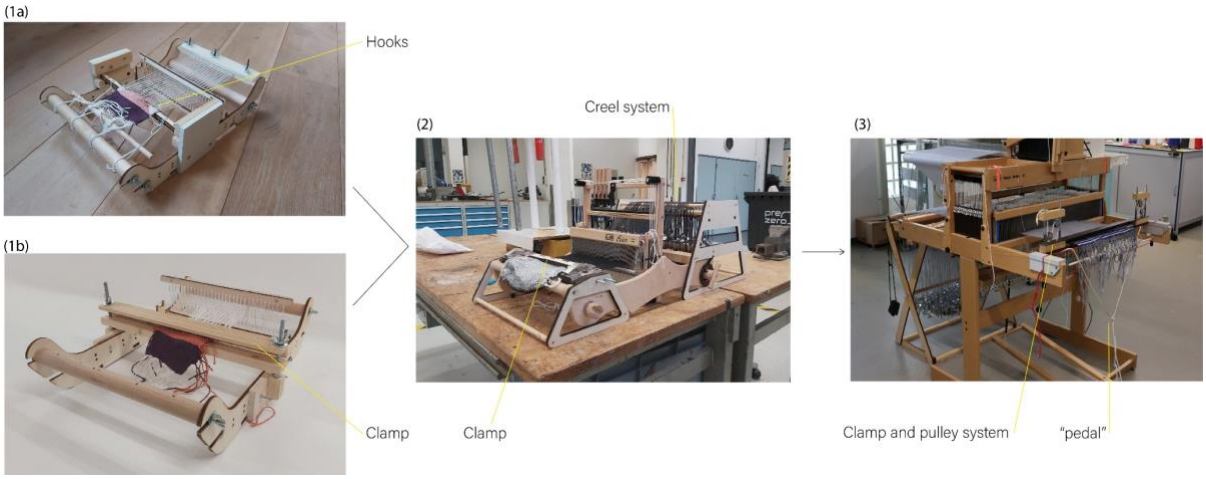


Figure 5: Prototypes made in the three phases of development of WarpForm. Pictures 1a and 1b correspond to the first iteration, which used low-fi and small-scale prototypes to quickly test the hook and the clamp. The second iteration (2) applied the clamp to a table loom and added a simple creel system. The third iteration (3) applied insights from iterations 1 and 2 to a dobby floor loom, and this set-up is what was used in this contribution.

3.1 Final WarpForm Prototype

WarpForm, in combination with a simple creel system, allows a variable warp consumption by using the reed to move the woven fabric past the fabric fell and locking the straightened fell in place with a clamp. The final prototype consists of two main parts: the fell clamp at the front of the loom and a simple creel system at the back (Fig. 6.a). Construction files for the prototype can be downloaded[1].

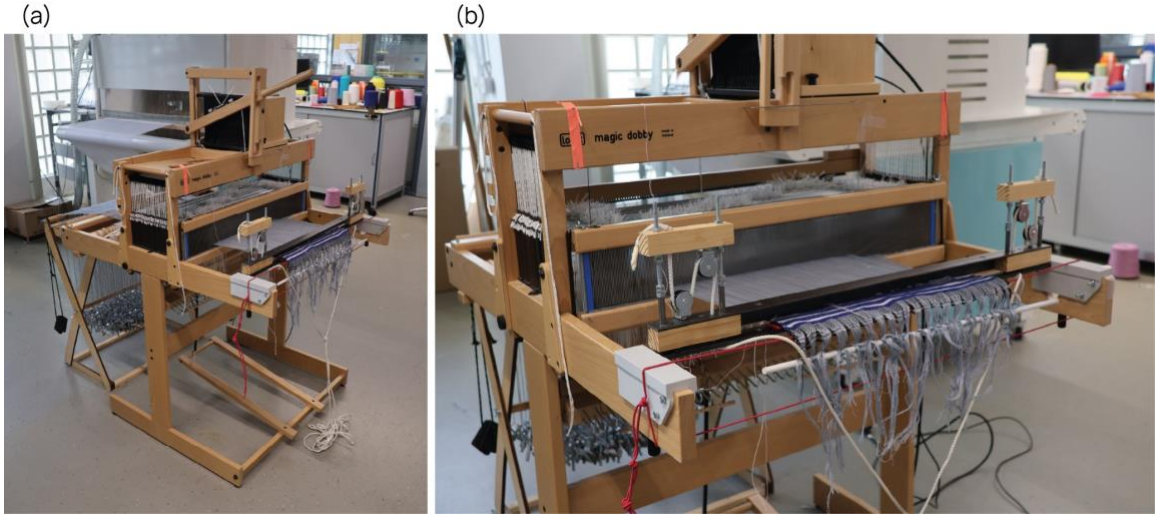


Figure 6: (a) The final WarpForm prototype with a fell clamp at the front of the loom and a creel system in the back. (b) The fell clamp at the front of the loom.

The main functionality of the fell clamp (Fig. 7) is to keep the warp in place at the fell so that the threads can be tensioned in the fabric-forming zone. The clamp consists of two steel halves with magnets on the bottom half to pull the top half toward it, securely holding the fabric in between. A layer of foam underneath the magnets compensates for slight thickness variations in the fabric.

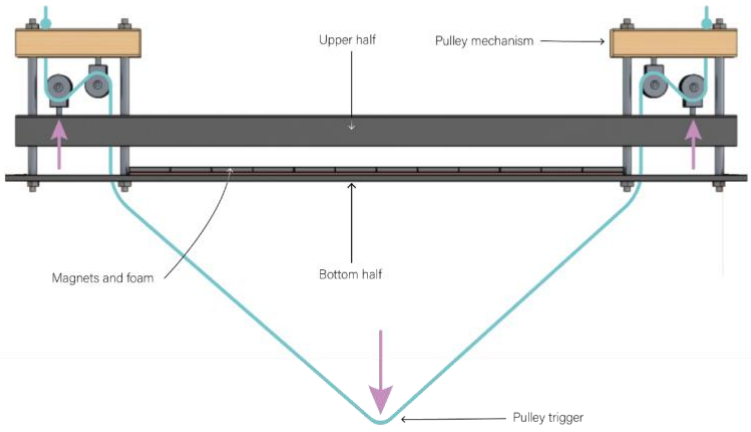


Figure 7. 3D model of the fell clamp showing the pulley mechanism that opens it.

[1] https://doi.org/10.4121/2f5f5c6d-402d-4e47-8b0b-a6888cfb2ef1

A pulley system, activated using the weaver's foot, opens the clamp. When the rope is pressed down, the upper part of the clamp is raised. When the rope is released, the upper part of the clamp falls down and grips the fabric. A pulley system reduces the effort required to lift the clamp's weight, and opening and closing the clamp with the foot leaves the hands free to move the reed forward and keep it in place, while aligning with the existing weaving control systems.

To attach the clamp to the loom, it is inserted into a slot in two 3D-printed brackets that clamp onto the loom's arms (Fig. 8.a). No permanent alterations to the loom are required, as the brackets are bolted into the existing holes of the loom's fabric beam.

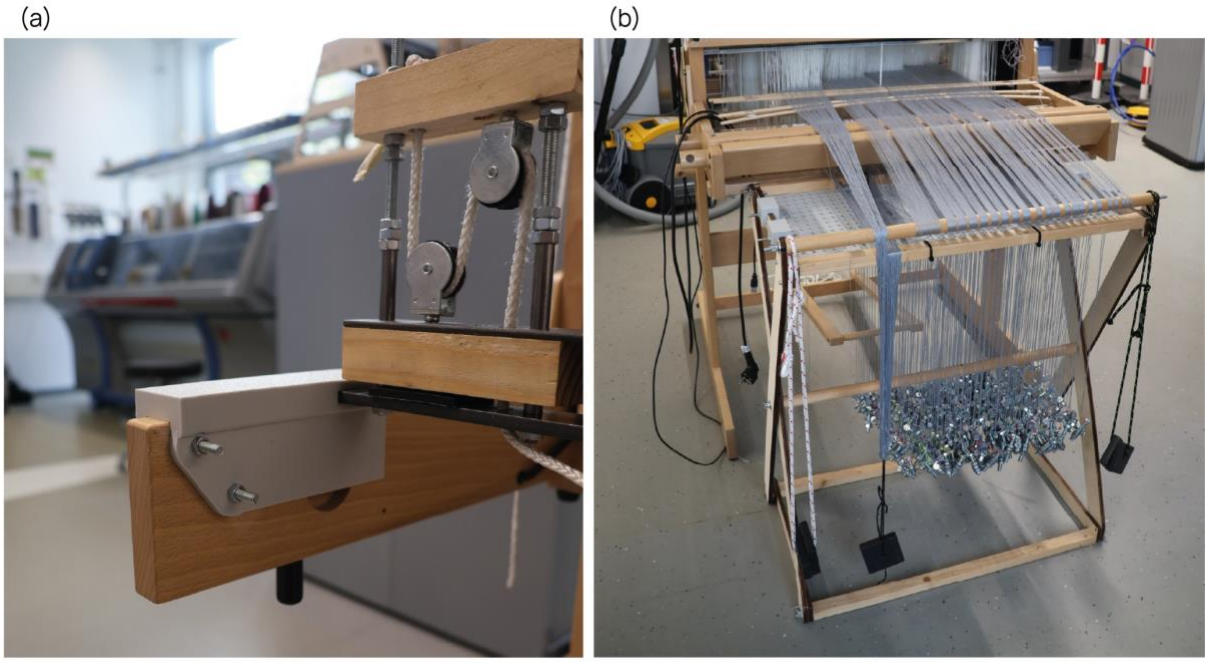


Figure 8: (a): Detail of the fell clamp attached to the loom with a 3D printed bracket. (b) Creel system at the back of the loom

WarpForm's fell clamp requires individual let-off of warp threads to function properly. Therefore, we built a simple creel system consisting of a let-off frame with a grid that creates space to hang weights on the warp threads to maintain the tension (Fig. 8.b). More information on this creel system can be found in Appendix A. Other creel systems could also be used in combination with WarpForm.

4. Weaving on WarpForm

4.1. Weaving Plan

Using WarpForm, material can be built up non-uniformly using short-row weaving, resulting in a non-rectangular or out-of-plane form, or both. Where and when partial weft rows are added during the weaving process to create the desired form are shown in a weaving plan - using a format derived from short-row knitting patterns.

Figure 9.a shows a weaving plan for a form consisting of two loops closed on one side (Fig. 9.c), created with rectangular areas of short row weaving in a plain 1x1 structure (indicated in pink on the weaving plan). With the reed and WarpForm, the woven sections are pushed together, closing over the bare warp threads (indicated in blue on the weaving plan) to form a fabric with a consistent density.

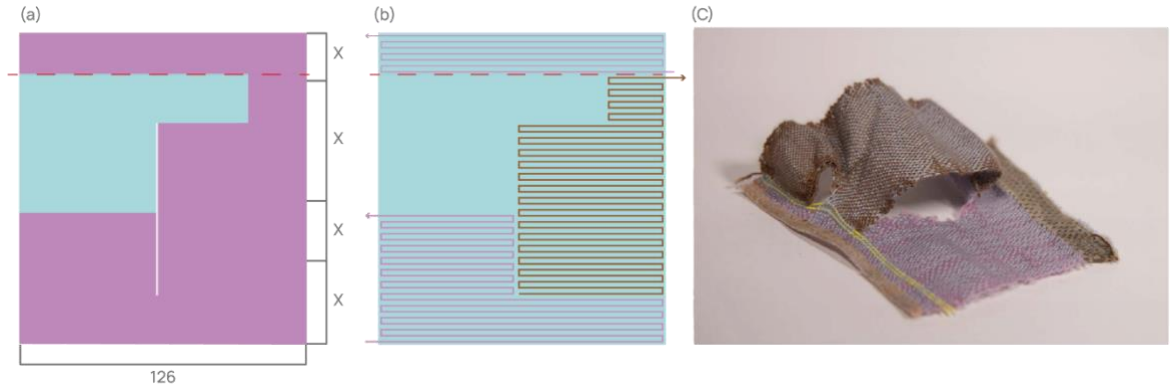


Figure 9: (a) Weaving plan: pink areas indicate areas where weft is interlaced with the warp (plain 1x1 structure), blue areas indicate empty warp threads. (b) Shuttle path: the way in which the weft travels through the warp based on the given weaving plan. (c) Picture of the resulting sample.

This sample also features a split area (indicated by a white line on the weaving plan), which requires two shuttles (differentiated by two colours in the weft shuttle diagram in Figure 9.b) so that the two adjacent areas remain separated. The process of weaving this sample is explained below.

4.2. Weaving Process

Weaving with WarpForm resembles that of a standard shuttle loom; shedding the warp, weft insertion, and beating are performed in the same manner. Short-row weaving techniques, in which the point where the weft enters and exits the warp is manipulated, also remain unchanged. A key difference, however, is that the fell is straightened with the reed at specific points to push the additional material created with partial weft rows out of the fabric-forming zone. How WarpForm is

used within the weaving process to create the above sample is shown in Figure 10. A video demonstrating the weaving process is also available[2].

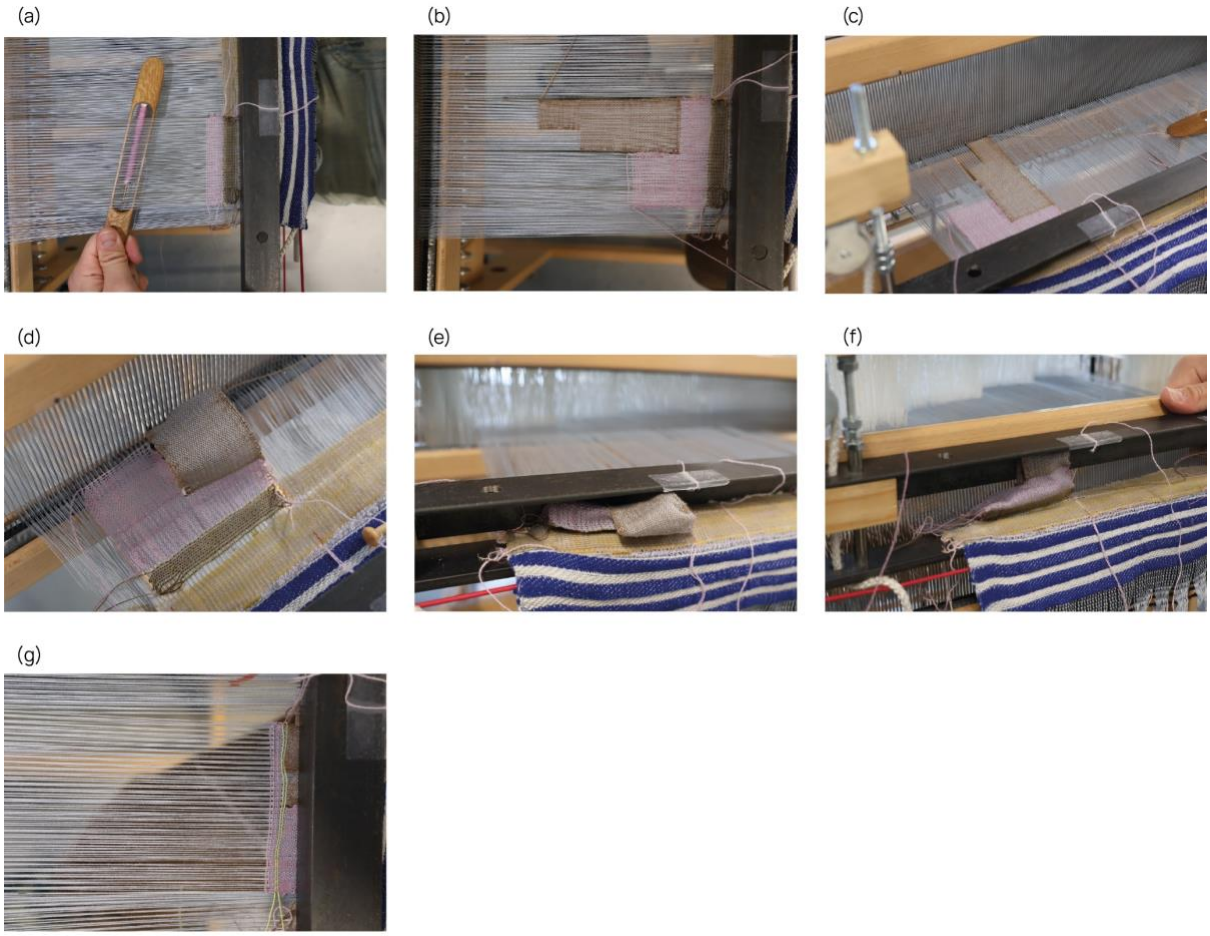


Figure 10: WarpForm weaving process for a sample with two loops.

The WarpForm weaving process is as follows:

1. A section of the warp is selected for the sample (Fig. 10.a).

2. Two step-like areas of partial weaving are woven (brown yarn) (Fig. 10.b) following the weaving plan.

3. The next desired weft row should be wider than the last row of the previously woven section (brown), requiring the fell to be straightened to allow the successive row to be beaten up neatly and leave no gaps (Fig. 10.c).

4. To straighten the fell, the reed is drawn toward the weaver and pressed against the last inserted weft row, while simultaneously opening the clamp using the foot pedal. The reed drags the weft

[2] https://doi.org/10.4121/2f5f5c6d-402d-4e47-8b0b-a6888cfb2ef1

and the interlaced warp threads towards the weaver, pushing the woven material to the fell and up into a loop (Fig. 10.d).

5. The reed, with the newly straightened fell resting against it, is pulled forward until it reaches the clamp, and the fell is positioned within the clamp opening (Fig. 10.e).

6. While holding the reed in place, the clamp is closed. The fabric behind the clamp is untensioned and starts to take on its 3D form (Fig. 10.f).

7. The straightened fell allows the next section to be woven neatly against the 3D form retained behind the clamp (Fig. 10.g).

It is worth noting that action is required only a limited number of times during the weaving process, namely whenever a wider row must be inserted. The clamp otherwise remains closed, keeping the weaving environment comparable to that of a standard hand loom.

4.3. Reflection on the weaving process

To gain insight into the practical realities of weaving with the final WarpForm prototype, a variety of experiments were conducted. Authors Savanne and Manuel, who wove most of the samples, observed that despite their limited weaving experience prior to this research, judging where to insert additional material to shape the fabric and where to straighten the fell was quite intuitive. However, the WarpForm prototype had some usability issues. Thickness variations in the fabric at the fell reduced the reliability of the clamp, and manual intervention was occasionally required to smooth the fabric to create a flat clamping area. Furthermore, because the warp threads were tensioned using weights, the weaver had to exert significant force to straighten the fell.

5. Samples

To understand and demonstrate how specific weaving decisions, aided by WarpForm, lead to a physical form, a series of weaving experiments was conducted on the dobby loom set up with 240 cotton warp threads with a density of 8 ends per cm. Cotton weft yarn was used for all the samples.

Each sample was created through short-row weaving in 1x1 plain weave, adding partial weft rows in specific areas to build up a form. To structure and communicate these possibilities, the resulting samples were organised into four form typologies (Fig. 11): *Soft forms*; *Sharp forms*; *Flat curves*; and *Warp sections*.

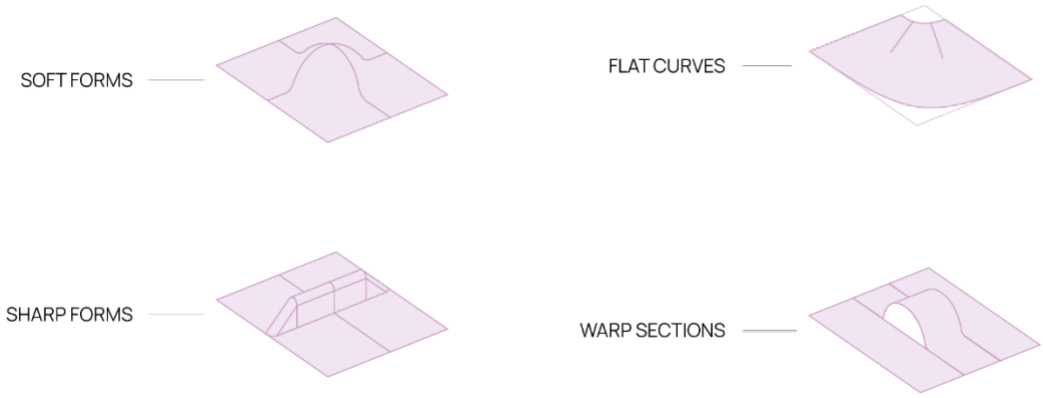


Figure 11: Overview of four form typologies and a categorising sample.

Each typology is expanded upon below, with one sample per category serving as an example of how a weaving plan relates to the final form.

5.1. Soft Forms

The *Soft forms* (Fig. 12) consist of organic shapes made by gradually adding material. Softly concave and convex curvatures are enabled, allowing the formation of outcomes such as domes.



Figure 12: Soft forms (a) Picture of a half sphere sample (Sample 1), (b) Drawing of the form. (c) Weaving plan, (d) Other views and samples of the soft forms cluster.

5.2. Sharp Forms

The Sharp form typology (Fig. 13) yields more precise, geometrically defined volumes. The resulting angular protrusions can produce nearly closed pockets and voids without sewing.



Figure 13: Sharp forms. (a) Picture of Ridge sample (Sample 3). (b) Drawing of the form. (c) Weaving plan ridge. (d) Other views of this sample and samples of this typology.

5.3. Flat Curves

The *Flat Curves* typology (Fig. 14) explores the non-rectangularity of the textile plane. In these, the warp and weft directions change as the textile is woven, allowing the textile's perimeter geometry to be shaped.



Figure 14: Flat curves, (a) Picture of Spiral sample (Sample 6). (b) Drawing of the form. (c) Weaving plan. (d) Other samples of this typology.

5.4. Warp Sections

The *Warp Sections* typology (Fig. 15) investigates splitting the warp into strips. This enables some sections to be woven while others are not, and can result in embedded loops, partial use of the warp (Fig. 15), or other nonrectangular textile outcomes.



Figure 15: Warp Section: (a) Picture of a single Loop sample, Warp system, Symmetrical. (b) Drawing of the form. (c) Weaving plan. (d) Other views and samples of this typology.

6. Demonstrator: shoe upper textile-form

To demonstrate a potential research application of WarpForm, and contextualise the woven samples and typologies, a one-seam shoe upper textile-form was developed as a demonstrator. Footwear was chosen as a domain because shoes are inherently form-sensitive objects: the upper must conform to the complex morphology of the human foot, often requiring cut-and-sew methods when using flat fabrics.

The prototype consists of a continuous woven piece of 186 warp threads and a density of approximately 16 ends per cm that wraps around the silhouette of a shoe last (Fig. 16). Cotton yarns were used for both the warp and weft. Woven entirely in plain weave, it comprises three distinct morphologies that together constitute the overall upper form, each corresponding to variations of samples from soft, sharp and flat curved forms. Respectively: two soft bulges to accommodate the heel, a sharp surface transition to cover the toe box (Fig. 16.a), and a flat curve repeated four times to shape the vamp (Fig. 16.b). As with the samples, partial weaving is also used to define the upper contour. To aid legibility, each instance of form-giving is marked with brown thread, and each fell straightening with yellow thread.

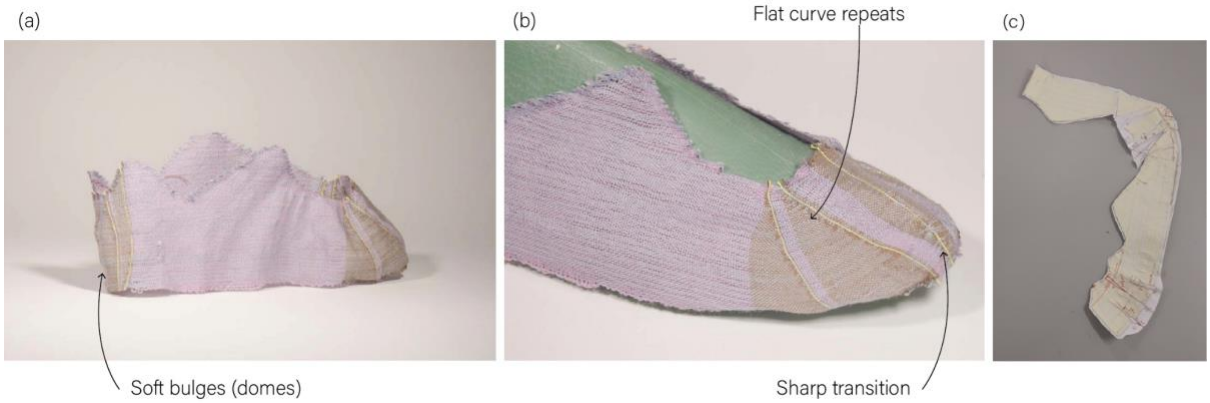


Figure 16: (a) back and side view of a shoe upper prototype; (b) front close up of the shoe upper prototype; (c) physical paper and tape pattern followed to weave the upper.

7. Discussion

Many weavers seek to push the boundaries of their craft, but are limited by their looms' capabilities. In this paper, we have presented WarpForm, an add-on for hand-operated dobby looms that enables the creation of 3D and non-rectangular Woven Textile-forms, and showcase its use through samples and a shoe upper prototype.

In the following sections, we discuss the limitations and future directions of this research. Through the hands-on making of the samples and the shoe prototype, we have identified the main usage challenges of the add-on and reflected on future improvements. WarpForm shows that a simple adjustment of a hand-operated loom can open up a whole new range of weaving possibilities. We hope this add-on lowers the barrier to experimenting with the creation of 3D and non-rectangular Textile-forms by enabling textile designers and researchers to use their making skills to imagine future directions towards sustainable textile systems.

7.1. Form potentials

The samples and prototype presented demonstrate that WarpForm opens new possibilities for creating 3D textiles, including domes, curves, loops, and corners. These samples provide a first glimpse of what WarpForm enables, but we believe its potential could go much further, particularly in combination with jacquard shedding. For example, the samples presented in this paper focused solely on using short-row weaving with a plain weave to shape the fabric. While short-row weaving enabled intuitive experimentation, resulting in a wide range of forms, future research should explore the potential of different weave structures, such as compound, multilayer, and other complex weave structures, to further support the creation of form and reduce post-weaving processes.

Furthermore, the samples created in this study did not fully exploit the potential of yarn as a material variable. Different yarn properties, such as stiffness, shrinkage, and melting, can be leveraged and combined with the techniques outlined in this paper to enhance the creative and technical possibilities of WarpForm woven textiles.

As the first users of WarpForm, we gained a first insight into how a weaver interacts with this tool and weaving process. However, user tests with other weavers of varying skill levels will be essential to test the comprehensibility of the weaving process, and work towards refining the tools and knowledge resources that could support weavers in exploring other possibilities.

7.2. Usage challenges

Through the hands-on exploration of WarpForm, some usage challenges emerged. While some were addressed during the tool's development, further improvements would enhance its user-friendliness and reliability. For example, rollers could be used to smooth the fabric before clamping, the clamp could be adjustable to accommodate thickness variations, or the hook system could be further developed. Furthermore, because there are many sizes and shapes of hand-operated looms, it would be important to make WarpForm compatible with or adjustable to different loom types, which may require a different clamp width, adjusted placement of the clamp to ensure alignment with the reed, and/or a different method of attaching the clamp to the loom. Additionally, looms set-up with a larger number of warp threads may require creel systems that are less reliant on the weaver's strength, for example by using electronic systems that automatically release the right amount of each warp thread or augment the weaver's pulling force.

The current version of WarpForm is a functional proof of concept. Further research should allow other weavers to user-test the current tool and future improvements. Improving the tool's usability will enable a wider range of users to experiment with WarpForm's potential, and we anticipate this will lead to uses for WarpForm beyond textile-form applications.

7.3. Future fabrication of Woven Textile-forms

Creating form directly on the loom presents a promising alternative to current wasteful cut-and-sew practices, as it enables weavers and designers to additively build textile objects. While researchers have explored Textile-form Weaving on both industrial and hand-operated jacquard looms (McQuillan, 2020; Piper & Townsend, 2015; Voorwinden & McQuillan, 2025; Shi et al., 2024), these studies also highlight the limitations of standardised machines, which are primarily designed for flat weaving. This, combined with the limited accessibility of specialised 3D weaving looms and tools, limits our understanding of how 3D and non-rectangular forms can be designed and fabricated.

WarpForm was developed to address this gap, providing an accessible, low-cost add-on that enables textile researchers and designers to experiment with non-rectangular and 3D woven textiles through hands-on prototyping. By modifying key loom components and supporting direct engagement with form, WarpForm opens up new possibilities for textile creation beyond flat surfaces.

Through hands-on experimentation and prototyping, designers gain insight into how material properties, techniques, and process variables interact, and explore how future fabrication methods might evolve. These tangible outcomes, such as samples and prototypes, not only make emerging ideas visible and tactile but also support more informed discussions with industry partners and machine manufacturers, particularly around desired outcomes and the feasibility of scaling such innovations. Expanding access to exploratory tools, such as WarpForm, allows a broader group of

practitioners to contribute insights that inform the development of future looms and production systems.

7.4. Conclusion

WarpForm, an add-on for a standard hand-operated loom introduced in this paper, was designed to be used in combination with a simple creel system to vary the consumption of individual warp threads, enabling the creation of 3D and non-rectangular forms. Using a clamp, the warp is fixed at the fell to maintain tension in the fabric forming zone while allowing the woven fabric to take on its 3D or curved shape. Aided by WarpForm, short-row weaving was used to create a variety of samples and a prototype of a shoe upper, indicating the tool's form potential.

With WarpForm, we aim to make the weaving of 3D and non-rectangular forms accessible to textile researchers and designers. We believe this accessibility is vital to enable weavers to gain unique insights through hands-on experimentation that could potentially push the textile industry in a more sustainable direction.

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Appendix A: Creel system

To create a creel system for use with WarpForm, a let-off frame was made over which the warp threads are led after leaving the warp beam (Fig. 1). The warp threads can still be stored on one or multiple warp beams. Usually, the threads would travel straight from the warp beam, over the back beam, and into the loom. The let-off frame was placed behind the loom and enabled the path of the warp threads to be lengthened. The warp threads travelled over the frame and around the back rod before travelling back towards the loom. In the space between the two beams of the let-off frame, the excess warp length that has been let off the warp beam was allocated and tensioned by hanging weights from them.

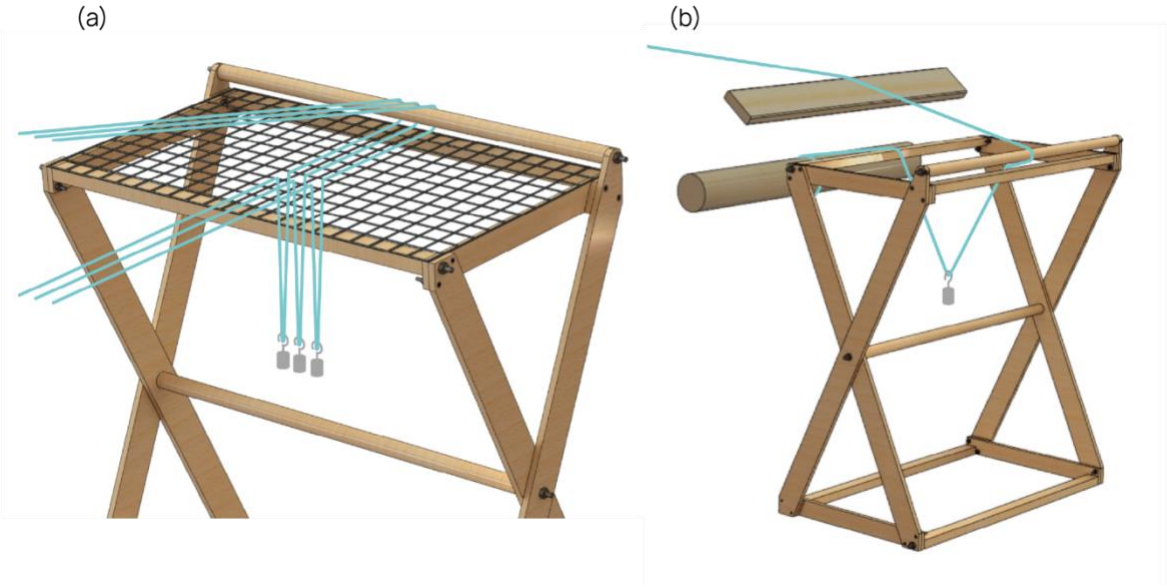


Figure 1: (a): Render illustrating how the grid separates the warp threads; (b): Render of the let-off frame with one warp thread.

To prevent the threads from tangling with each other, a grid was added to the let-off frame (Fig. 1a). The warp threads were divided over the grid holes to keep them separated.

In the set-up used for the samples, threads were tensioned with 30gr weights, which allowed each warp thread to be pulled forward individually, while ensuring any excess length was retracted to bring the warp thread back to the right tension (Fig. 2).

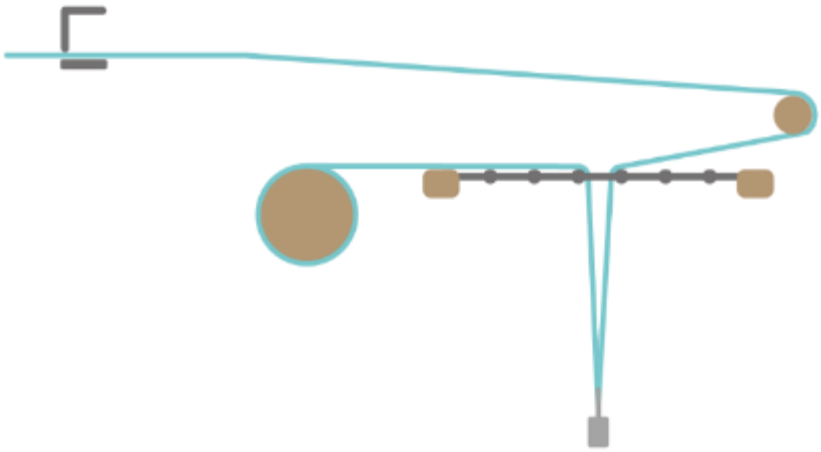


Figure 2: Schematic showing the path of the warp thread, from the let-off beam to the clamp.

As for the demonstrator configuration, every two warps were tensioned with a weight of 35 gr. To balance the force exerted by the weaver when pulling the warp threads forward, counterweights were added to the back of the let-off frame (Fig. 3).



Figure 3. Creel system with counterweights hanging on the back of the let-off frame.